

TECHNICAL SPECIFICATION FOR HIGH VOLTAGE DETECTOR

1.0 The High Voltage Detector (HVD) shall be suitable for detection of liveness of electrical infrastructure including overhead bare conductor distribution lines. The HVD shall start annunciation with flashing Red LED display and buzzer beeps indicating that the line / infrastructure is LIVE and not safe to earth and/or carry out any operation. Sensing distance shall have to be at least 3 M for 33 kV and 1 M for 11 kV lines and 5" for 415 V lines.

2.0 SERVICE CONDITION:

The High Voltage Detector to be supplied as per this specification shall be capable of detecting live lines on bare conductors under hot, tropical and dusty climate and shall be suitable for satisfactory operation under the following tropical conditions.

- Maximum Ambient Air Temperature in shade: 50°C
- Minimum Ambient Air Temperature: (-) 5°C
- Maximum Relative Humidity : 95 %(non- condensing)
- Minimum Relative Humidity: 10%
- Maximum Rainfall: 2000 mm
- Maximum wind Pressure: 150Kg/Sq.mm
- Climatic condition: Moderately hot & humid tropical climate

3.0 SYSTEM CONDITIONS:

The High Voltage Detector is intended for use in LT & HT Distribution System. Distribution System is 1Φ2W / 3Φ3W / 3Φ4W and has the following data:

System Voltage Level	Nominal System Voltage	33 kV	11 kV	415 V
	High System Voltage	36 kV	12 kV	440 V
	Line to Earth Voltage(Nominal)	19.05 kV	6.35 kV	230 V
Insulation Level (Lightning Impulse Voltage)		170 kV	95 kV	1.1 kV
Power Frequency withstand Voltage		70 kV	28 kV	NA
Rated Frequency		50 Hz		
Insulation Resistance		Greater than 100MΩ		
Sound Pressure		90dB/Metre ± 10dB/Metre		

4.0 POWER SUPPLY:

- Dry Battery with suitable Voltage Ratings
- Commercially available everywhere

- 7.12 The silicon petticoat arrangement shall be fixed on rod for decrease of the flash over contact probability with the operator.
- 7.13 The detector shall not have any external accessibility to change the voltage or the sensing distance to ensure the safety of a user to avoid confusion at the site.

8.0 DETECTION INDICATION:

- 8.1 The High Voltage Detector shall give the indication with High Intensity LEDs with flashing arrangement there by providing the suitable indication in day light.
- 8.2 The High Voltage Detector shall give audible sound for the presence of live line so that it can be suitably used in populated and noisy areas for easy operation.

9.0 DIMENSIONS:

- 9.1 The High Voltage Detector probe shall have universal connecting link with suitable length for the attachment of the Telescopic Rod.
- 9.2 Length of the Telescopic Rod shall be at least 4800 mm.
- 9.3 The detachable type Telescopic Rod shall have four sections/pieces, each section/piece having maximum length of 1600 mm & minimum length of 1200 mm (including coupling) with the bottom section having locking arrangement, to reach the required distance / height.
- 9.4 The outer diameter of the bottom part of the Telescopic Rod shall not exceed 45 mm with tolerance as per IS.

10.0 SUITABLE CARRYING CASE:

The High Voltage Detector with accessories shall be supplied with the carrying case so as to be carried by the operating person easily & individually.